

T14

$$X \sim N(a; \sigma_x^2)$$

$$\sigma_x^2 = 2$$

$$Y \sim N(b; \sigma_y^2)$$

$$\sigma_y^2 = 1$$

$$H_0: a = b$$

$$H_1: a > b$$

$$\Delta = \bar{x} - a - (\bar{y} - b) \sim N(0, \frac{2}{n} + \frac{1}{m})$$

$$H_0 - \text{верна} \Rightarrow \Delta = \bar{x} - \bar{y} \sim N(0, \frac{2}{n} + \frac{1}{m})$$

$$\text{Рассм. } G = \{ \Delta \geq u_{1-\alpha} \}$$

$$\text{Пусть } a - b = \delta$$

$$\tilde{\Delta} = \bar{x} - \bar{y} - \delta \sim N(0, \frac{2}{n} + \frac{1}{m}) \quad \text{т.е.}$$

$$\bar{x} - \bar{y} \sim N(\delta, \frac{2}{n} + \frac{1}{m})$$

$$\text{т.е. } W(\delta) = P(\tilde{\Delta} \geq u_{1-\alpha})$$